

Data scientist with 3+ years of experience in big data, natural language processing, supervised and unsupervised learning, deep learning, and deploying machine learning powered web-applications.

TECHNICAL SKILLS

Languages/Frameworks: Python, SQL, MongoDB, Flask, Django, AWS, PySpark

Machine Learning toolkits: TensorFlow, Keras, scikit-learn, XGBoost, NumPy, pandas, NLTK, SpaCy, gensim

Data Visualization Toolkits: Matplotlib, seaborn, bokeh, plotly

EXPERIENCE

Data Scientist **San Francisco State University** **November 2019- August 2020**

Opioid Addiction-Stage Analysis and Prediction on social media - ASAP (<http://haddock9.sfsu.edu/asap>)

- Performed time-series and NLP modeling on Reddit posts of opioid users to predict their stage of addiction (*use, withdrawal and recovery*).
- Implemented pointwise mutual information to determine stage specific n -grams. The word2vec embeddings of these n -grams were combined with other addiction specific information (extracted using regular expressions) to create feature vectors.
- These feature vectors were used to train a Bidirectional-LSTM classifier using Keras to classify posts into multiple stages of addiction. Achieved state of the art $F-1$ score of 88%. Currently under review in the Bioinformatics Journal.
- Developed the open-source web application "ASAP" using Python 3, Flask, MongoDB, jQuery, JavaScript, and Redis (in-progress).
- ASAP supports visualizing topic models, sentiment scores, histograms, and scatter plots for linguistic analysis of user posts.

Drug Use Insight - DUI (<http://haddock9.sfsu.edu/insight>)

- Created the first ever substance use and recovery taxonomy consisting of 108 categories and over 9,000 terms.
- 7 million posts made by 1.5 million users were analyzed using thematic analysis and word2vec to create the taxonomy.
- Co-designed and co-built the open-source web application "DUI" using Python 3, Django, SQL, CSS, bootstrap, and JavaScript.
- Given the social media posts, DUI can be used to extract constituent terms associated with substance use and recovery.
- DUI also provides several visualizations – histograms, pie charts, time series, and correlation plot for sensemaking.

Graduate Research Assistant **San Francisco State University** **January 2016 – October 2019**

Social Media based Drug Addiction Recovery Intervention Targeting System – SMARTS (<http://haddock9.sfsu.edu/>)

- Identified different drug use and recovery forums on social media platform Reddit using Bi-clustering and tf-idf.
- Implemented propensity score matching to identify n -grams that indicate an increase in the probability of an individual to post in an addiction recovery forum.
- Trained a Logistic Regression model to predict addiction recovery with an $F-1$ score of 87 % on an imbalanced dataset.
- Developed an open-source web application "SMARTS" by utilizing Python, Flask, jQuery, JavaScript, and Redis.
- SMARTS also presents visualizations to perform further addiction analysis. Published in the Bioinformatics journal.

Modeling the spread of HIV infection by identifying and characterizing high risk communities on social media.

- Tweets of users were used to train an SVM classifier to identify HIV related accounts on Twitter with an $F-1$ score of 78%.
- The followers of these accounts were grouped according to their geolocation to create location-based contact networks.
- Applied Louvain community detection on the contact networks to identify at-risk communities in New York and Illinois.
- Published in ISBRA and corroborated the findings by CDC that individuals involved in injection drug use are at-risk for HIV.

EDUCATION

San Francisco, CA **San Francisco State University** **August 2015 – November 2019**

- M.S. in Computer Science.
- Thesis: Modeling Addiction and Disease Epidemiology using Social Media.

Patiala, India **Thapar University** **August 2011 – August 2015**

- B.E. in Computer Science.

PUBLICATIONS

- Jha, D., & Singh, R. (2019). SMARTS: the social media-based addiction recovery and intervention targeting server. *Bioinformatics*. (Impact Factor – 4.5) 📄
- Skums, P., Zelikovskiy, A., Singh, R., Gussler, W., Dimitrova, Z., Knyazev, S., Mandric, I., Ramachandran, S., Campo, D., Jha, D. and Bunimovich, L., 2018. QUENTIN: reconstruction of disease transmissions from viral quasispecies genomic data. *Bioinformatics*, 34(1), pp.163-170. (Impact Factor – 4.5) 📄
- Jha, D., Skums, P., Zelikovskiy, A., Khudiyakov, Y., & Singh, R. (2017, May). Modeling the Spread of HIV and HCV Infections Based on Identification and Characterization of High-Risk Communities Using Social Media. In *International Symposium on Bioinformatics Research and Applications* (pp. 425-430). Springer, Cham. (Acceptance % - 20.6%) 📄
- Eshleman, R., Jha, D., & Singh, R. (2017, November). Identifying individuals amenable to drug recovery interventions through computational analysis of addiction content in social media. In *2017 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 849-854). IEEE. (Acceptance % - 19.9%) 📄